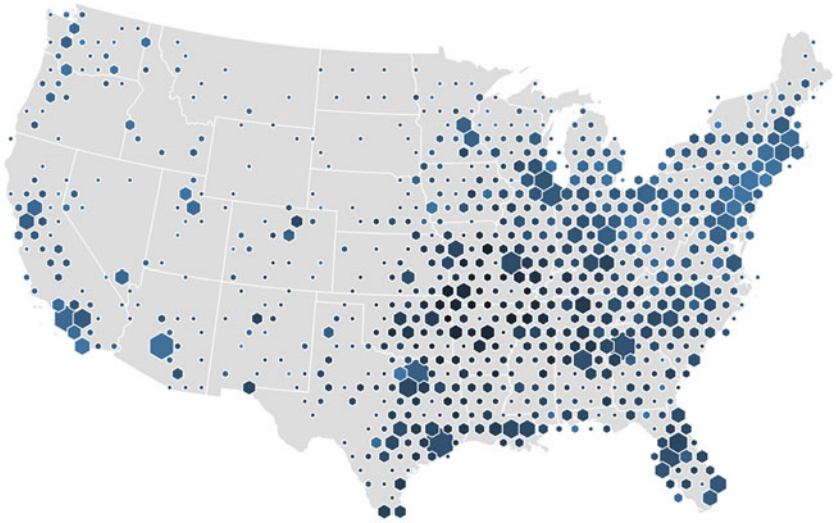


Figure 7.22 An example of hexbinning by Mike Bostock showing the locations of WalMart stores in the United States (available at http://bl.ocks.org/mbostock/4330486).

7.6.3 Raster reprojection

Jason Davies and Mike Bostock have both provided examples of reprojecting, not just vector data, but the tile data used in tile maps (see bl.ocks.org/mbostock/ and www.jasondavies.com/maps/raster/satellite/). You can use this to show a satellite-projected terrain map, or a terrain map with the Mollweide projection we used earlier.

7.6.4 Hexbins

The `d3.hexbin` plugin allows you to easily create hexbin overlays for your maps like that seen in figure 7.22. This can be effective when you have quantitative data in point form and you want to aggregate it by area.

7.6.5 Voronoi diagrams

As with hexbins, if you only have point data and want to create area data from it, you can use the `d3.geom.voronoi` function to derive polygons from points like the kind seen in figure 7.23.

7.6.6 Cartograms

Distorting the area or length of a geographic object to show other information creates a *cartogram*. For example, you could distort the streets of your city based on the time it takes to drive along them, or make the size of countries on a world map bulge or shrink based on population. Although no simple functions exist to create cartograms, examples of how to create them in D3 include one created by Jason Davies (http://www.jasondavies.com/maps/dorling-world/), one created by Mike Bostock (http://bl.ocks.org/mbostock/4055908), and the cost cartogram I built (orbis.stanford.edu).